Computer Networks (ICT Skills)

Computing and Mathematics

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Computer Networks (ICT Skills) (A13487)

Short Title: Computer Networks (ICT Skills)
Faculty Science and Computing
Department Computing and Mathematics
Cluster Networks and Cloud
Author LBWHITE
Credits 5 **Level:** Advanced

Description of Module / Aims

This module introduces data communications terminology and concepts, network protocols and models. Students will use protocol analysis software to explore various network protocol operations. An examination of TCP/IP, IP addressing and Ethernet is presented as well as a brief introduction to Routing and Wireless LANs. Practical skills are an essential part of this module.

Programmes

`COMP-0519`	Higher Diploma in Science in Computer Science (`WD_KCOSC_G`)	4 / 1 / M

Indicative Content

- Introduction to Computer Networks and Protocols
- OSI and TCP/IP models
- Ethernet
- Network Layer Protocols and Functionality
- IP Addressing and subnetting
- Routing
- Transport Layer Protocols and Functionality
- Application Layer Protocols and Functionality e.g. HTTP, FTP, DNS, SMTP
- Wireless LANs

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Compare network protocol models to explain the layers of communications in data networks.
2. Evaluate the major components, operation and functionality of a computer network and commonly used protocols and services.
3. Design, calculate and apply subnet masks and addresses.
4. Build a simple Ethernet network using routers and switches.
5. Configure routers and switches using command line interface.
6. Determine the operations and features of network protocols and services using protocol inspection software.
7. Implement a basic wireless network.

Learning and Teaching Methods

- The lectures will introduce the theory content to the student. The student will be encouraged to participate in class discussions and ask questions to support their learning process.
- The practical classes facilitate the student in implementing the theory learned in the lectures.
- The continuous assessment will require the student to apply the theory and practical knowledge to a business solution.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,2,3
Continuous Assessment	50%	
In-Class Assessment	50%	3,4,5,6,7

Assessment Criteria

<40%: Unable to describe the major functions and operation of a Computer Network. Unable to describe and compare the OSI and TCP/IP models. Poor understanding of role of communications protocols in computer networks.

40%-49%: Can describe and compare the OSI and TCP/IP models. Can provide overview of main computer network components and protocols.

50%-59%: All of the above. Can describe in detail the data encapsulation process. Demonstrate an understanding of basic LAN implementation.

60%-69%: In addition, be able to recommend a network solution given an organisations' requirements.

70%-100%: All the above to an excellent level. Be able to analyse and design solutions to a high standard for a range of both complex and unforeseen problems through the use and modification of appropriate skills and tools.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	12	
Practical	36	
Independent Learning	87	

Supplementary Material

- "Association for Computing Machinery." http://www.acm.org
- "Cisco." http://www.cisco.com/web/learning/netacad/index.html
- "IEEE Communications Society." http://www.comsoc.org
- "IEEE Computer Society." http://www.computer.org
- Dye, M., R. McDonald and A. Rufi. _Network Fundamentals: CCNA Exploration Companion Guide (Cisco Networking Academy)_. New York: Cisco Press, 2011.
- Tanenbaum, A. _Computer Networks_. 4th Ed.. New York: Prentice Hall, 2002.

Requested Resources

- Room Type: Computer Lab